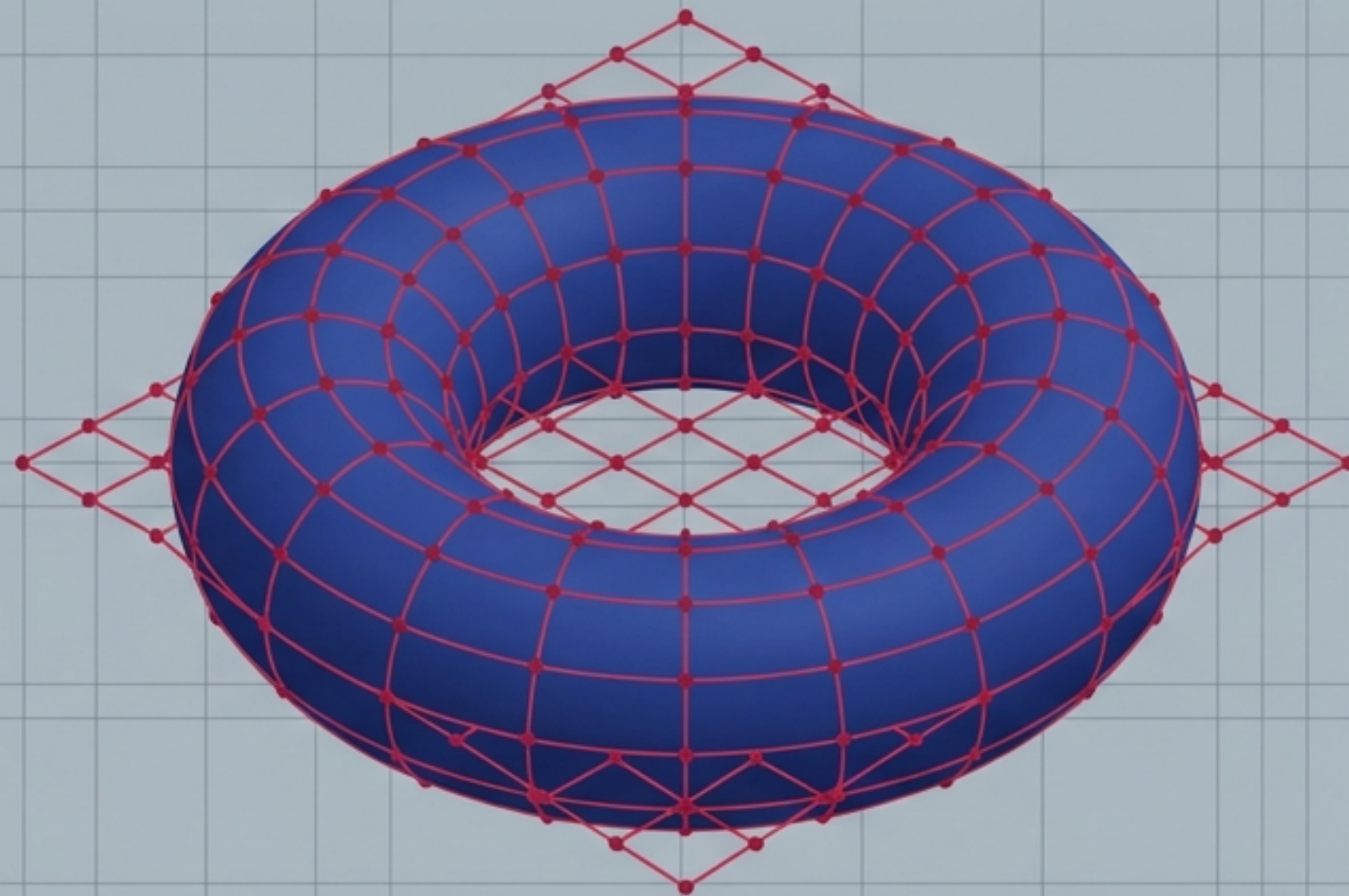


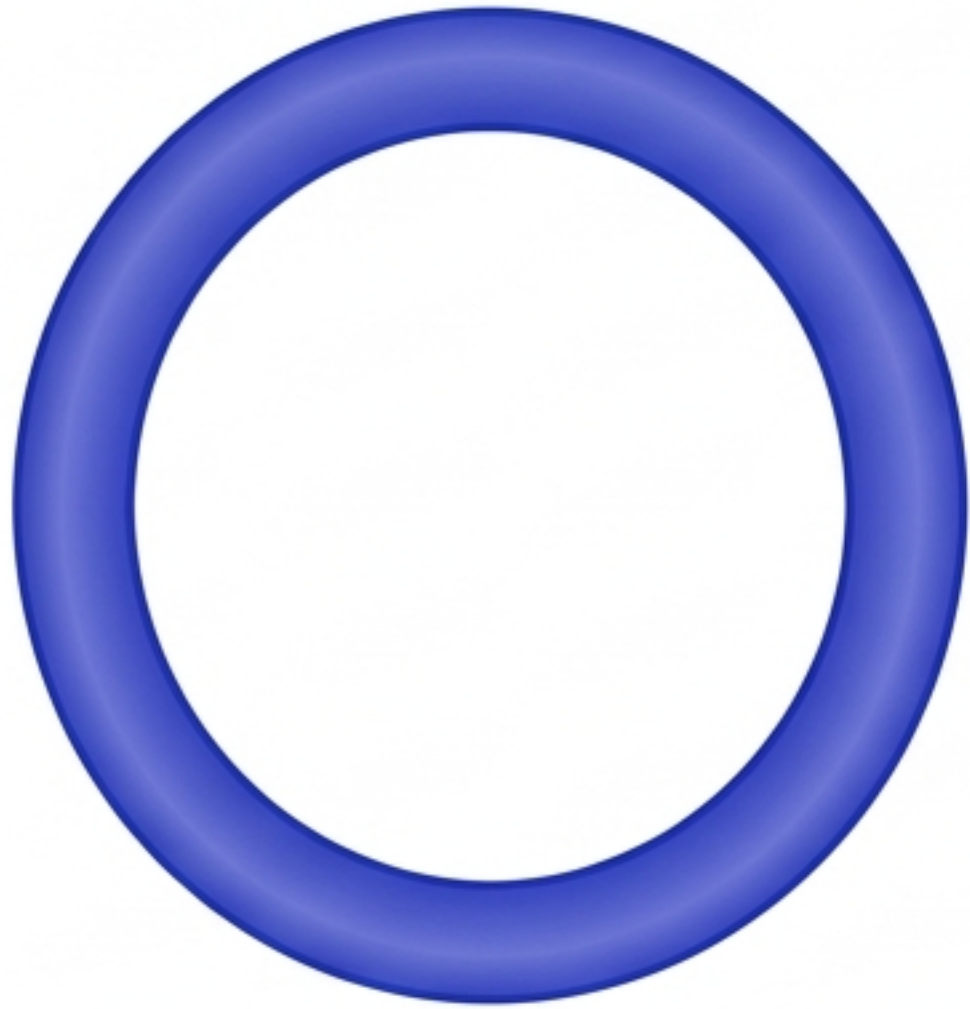
Solving the Hilbert-Smith Conjecture in Every Finite Dimension

A visual deconstruction of the
breakthrough September 2026
OpenAI preprint.

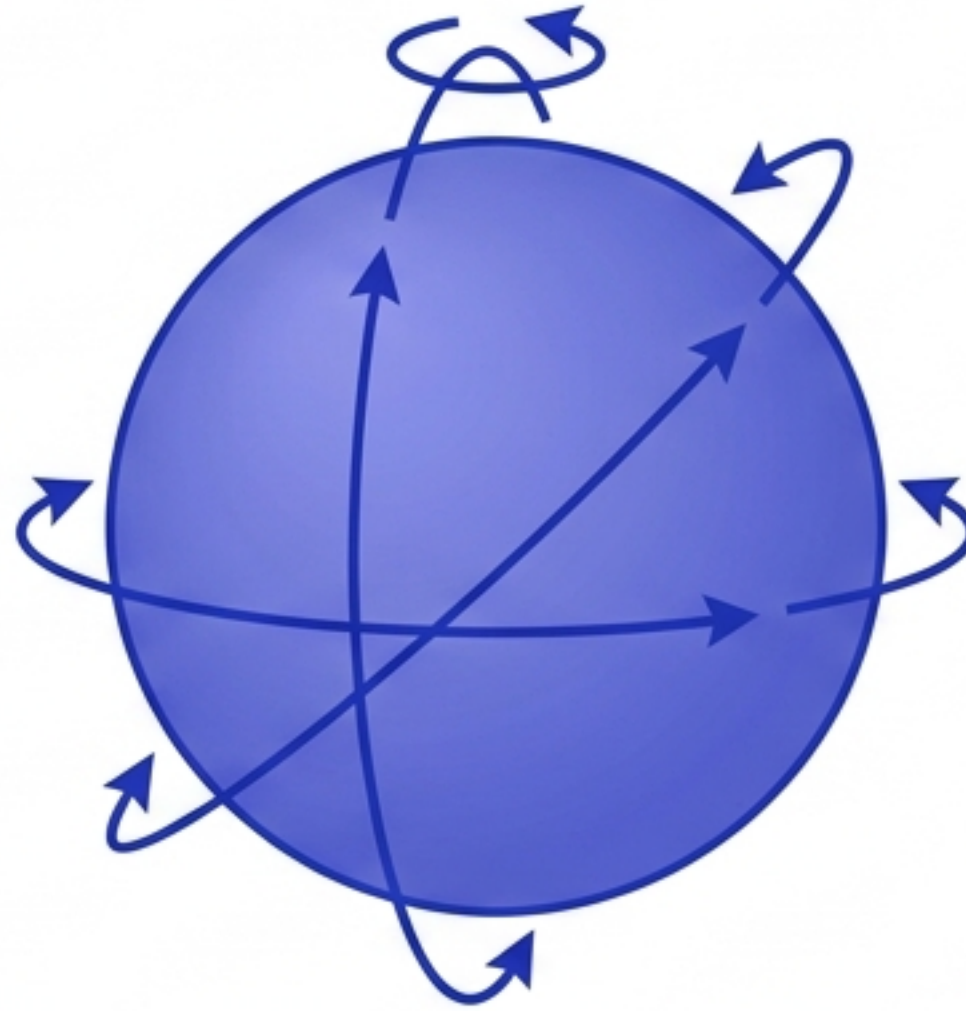


Lie groups are the mathematical embodiment of smooth, continuous symmetry.

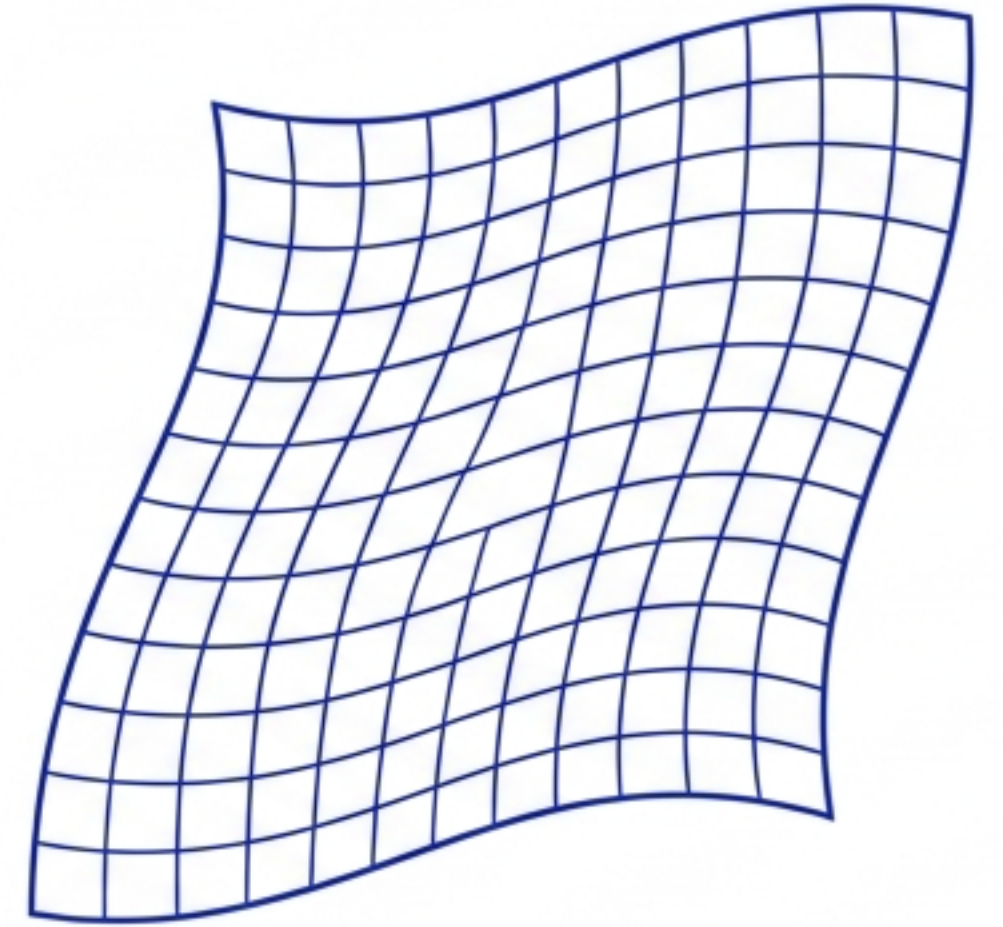
A Lie group is a group that is also a smooth manifold. It allows us to apply the tools of calculus to concepts of symmetry.



S^1
(1D rotation)



$SO(3)$
(3D rotation)



$GL_n(\mathbb{R})$
(Invertible matrices)

Hilbert's Fifth Problem asked if topological assumptions alone force a smooth structure.



The 1900 Mystery

“If a topological group is a manifold, must it be a Lie group?”

1952 Resolution

Gleason and Montgomery-Zippin prove the answer is YES for groups that are manifolds.

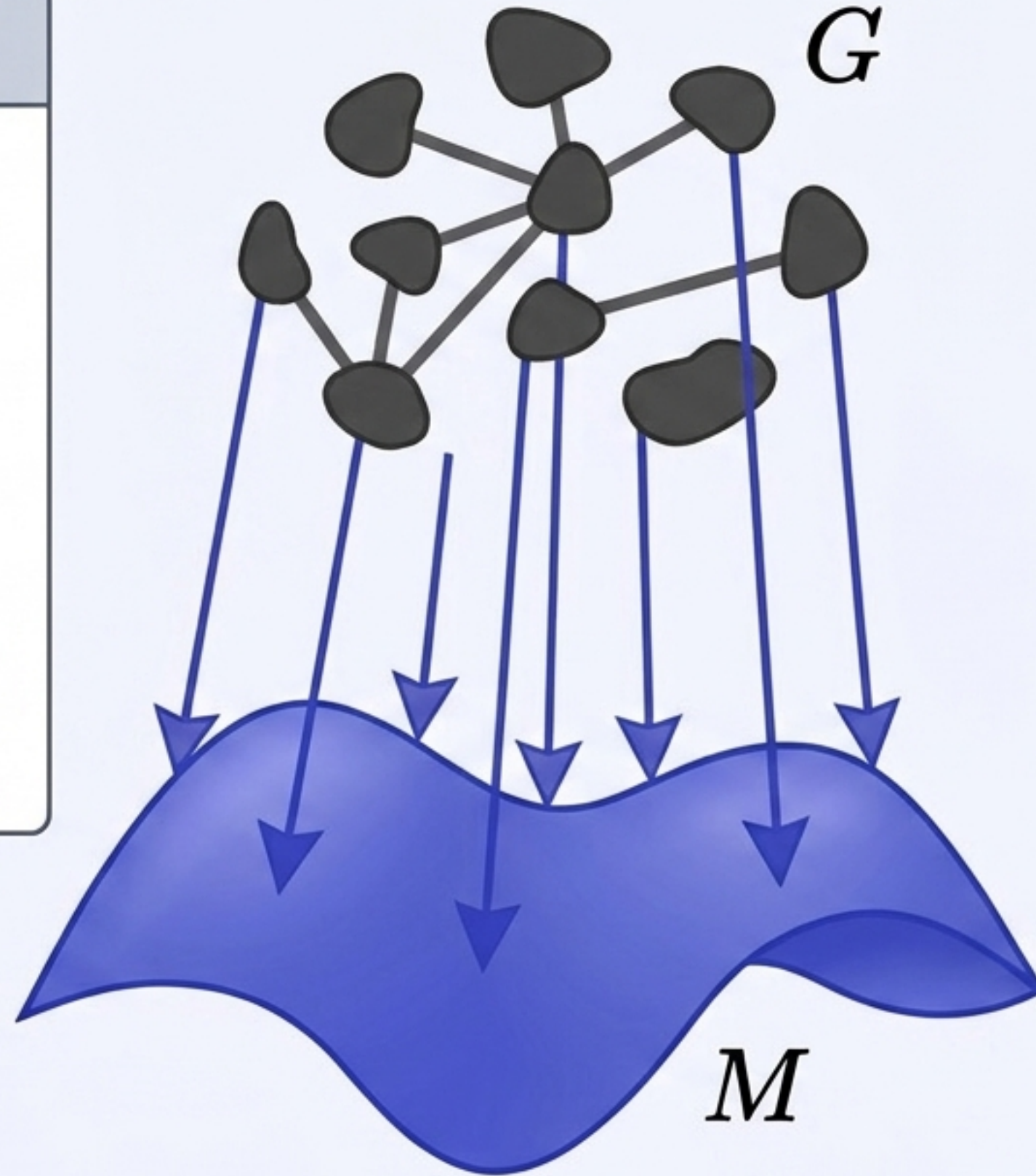
1953 Extension

Yamabe extends the structure theory to general locally compact groups.

The Hilbert-Smith Conjecture pushes the question to transformation groups.

The Conjecture

If a locally compact group acts faithfully and continuously on a connected topological manifold, it must be a Lie group.



Continuous

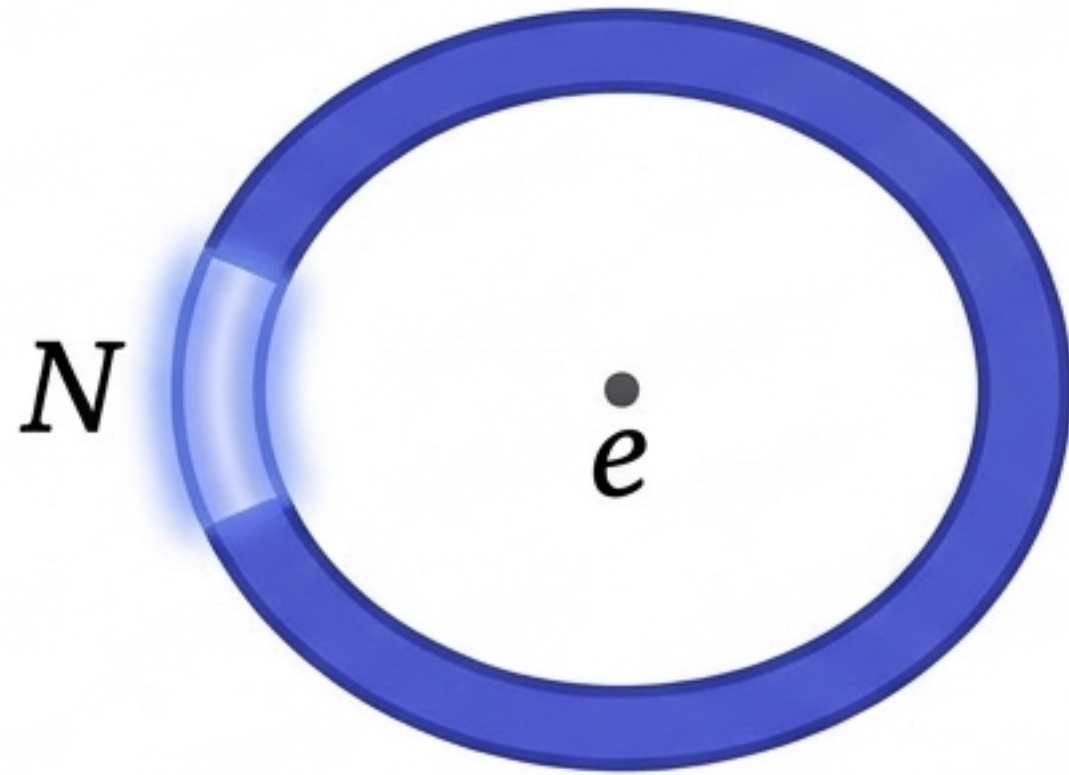
The transformation doesn't tear the manifold.

Faithful

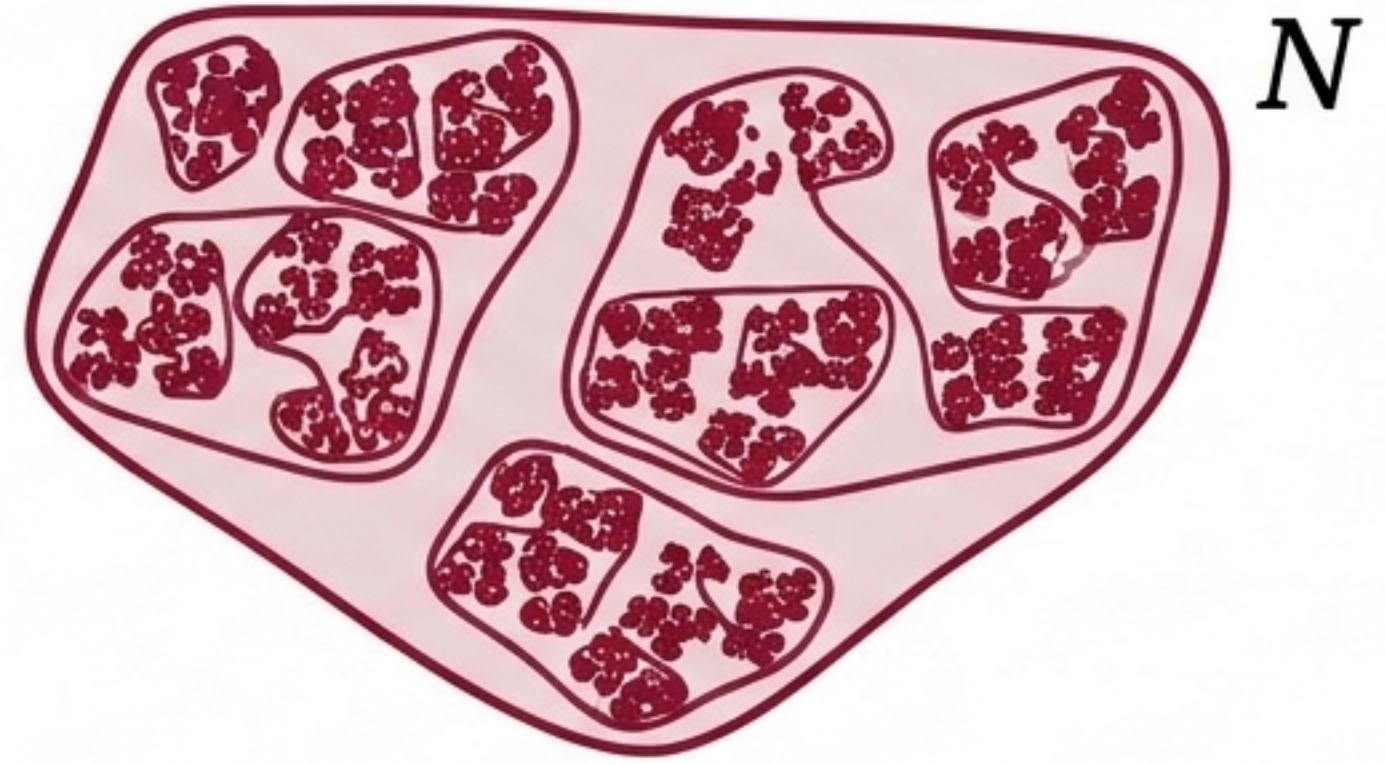
No element (except the identity) leaves every single point unmoved.

The diagnostic litmus test for a Lie group is the absence of small subgroups.

Circle Group (S^1)



p -adic Integers ($\mathbb{Z}_p$) Uncaltrured



Small Subgroups

NONE. A small arc around e
contains no full subgroups.
It is a Lie group.

Small Subgroups

INFINITE. Any neighborhood
contains nested subgroups $p^k \mathbb{Z}_p$.
Not a Lie group.

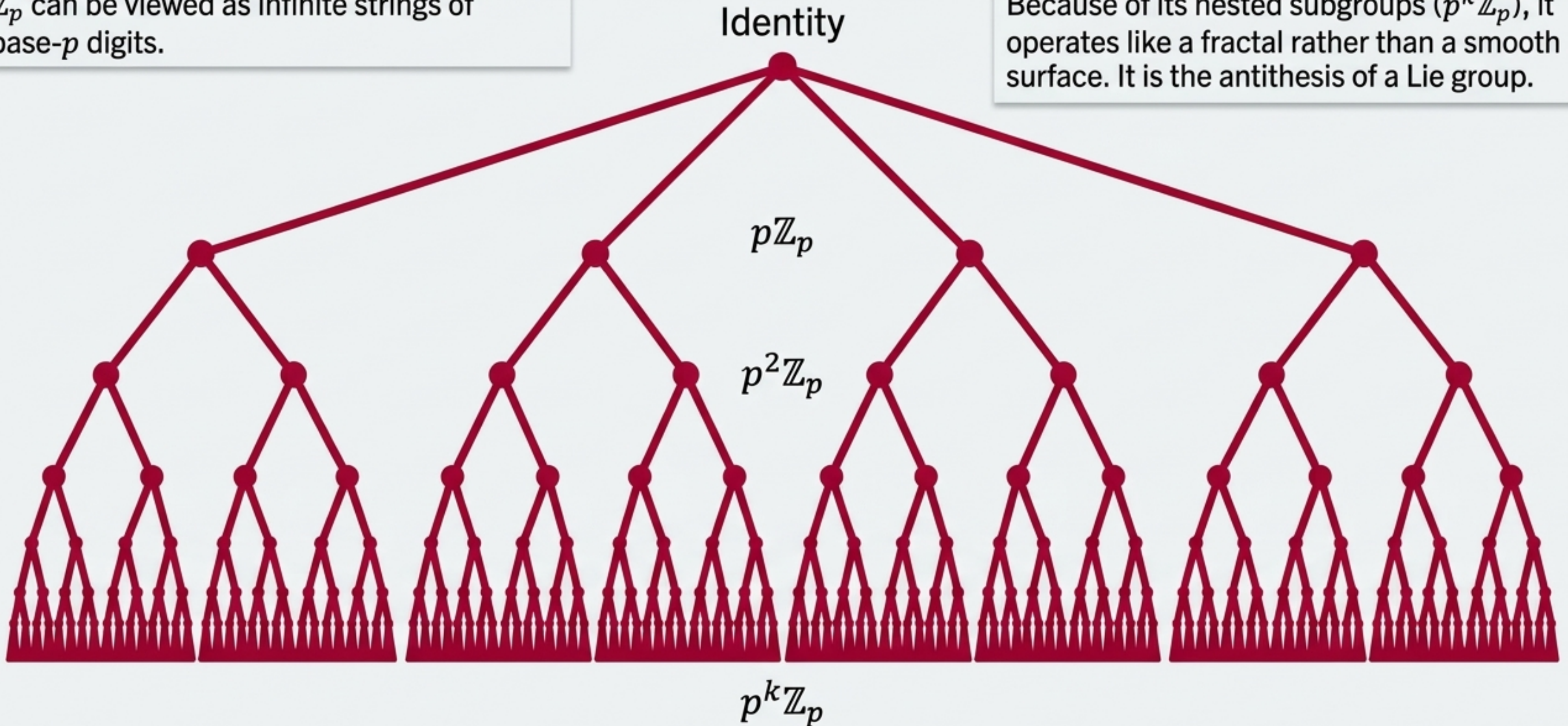
The p -adic integers form a totally disconnected, infinitely branching hierarchy.

Structure

$\mathbb{Z}_p$ can be viewed as infinite strings of base- p digits.

The Obstruction

Because of its nested subgroups ($p^k\mathbb{Z}_p$), it operates like a fractal rather than a smooth surface. It is the antithesis of a Lie group.



The conjecture reduces entirely to proving that $\mathbb{Z}_p$ cannot act faithfully.

All Locally Compact Groups

The Classical Reduction

By mid-century structure theory, proving the Hilbert-Smith conjecture requires exactly one thing: showing the additive group of p -adic p -adic integers ($\mathbb{Z}_p$) cannot act faithfully on a manifold.

Montgomery-Zippin /
Yamabe Structure Theory

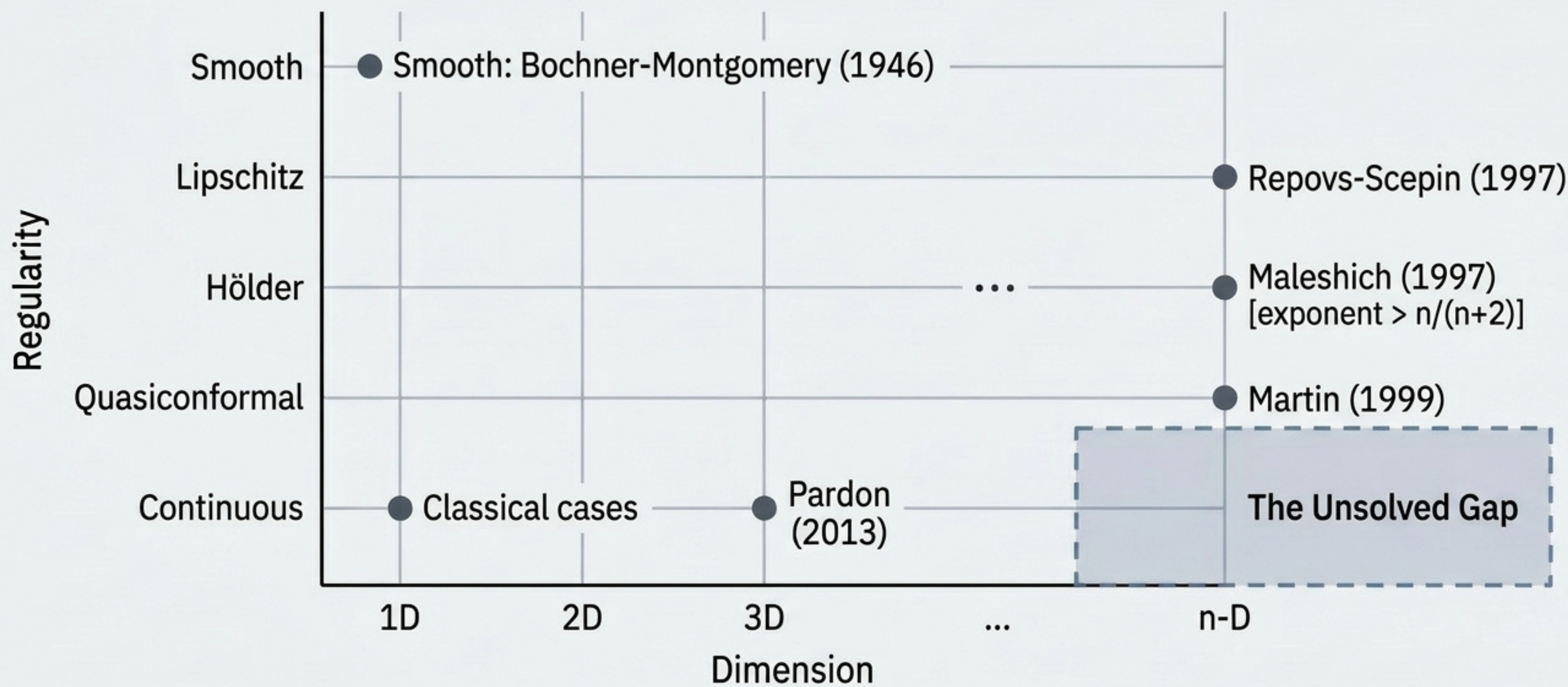
The Local Tool

Newman's 1931 Theorem dictates that a finite-order homeomorphism fixing an open set pointwise must be the identity.



$\mathbb{Z}_p$ acting on a manifold

A century of partial solutions relied heavily on low dimensions or forced geometric regularity.



Dimension Penalties: Yang's orbit-space dimension $n+2$.

The September 2026 OpenAI preprint claims a full topological resolution.

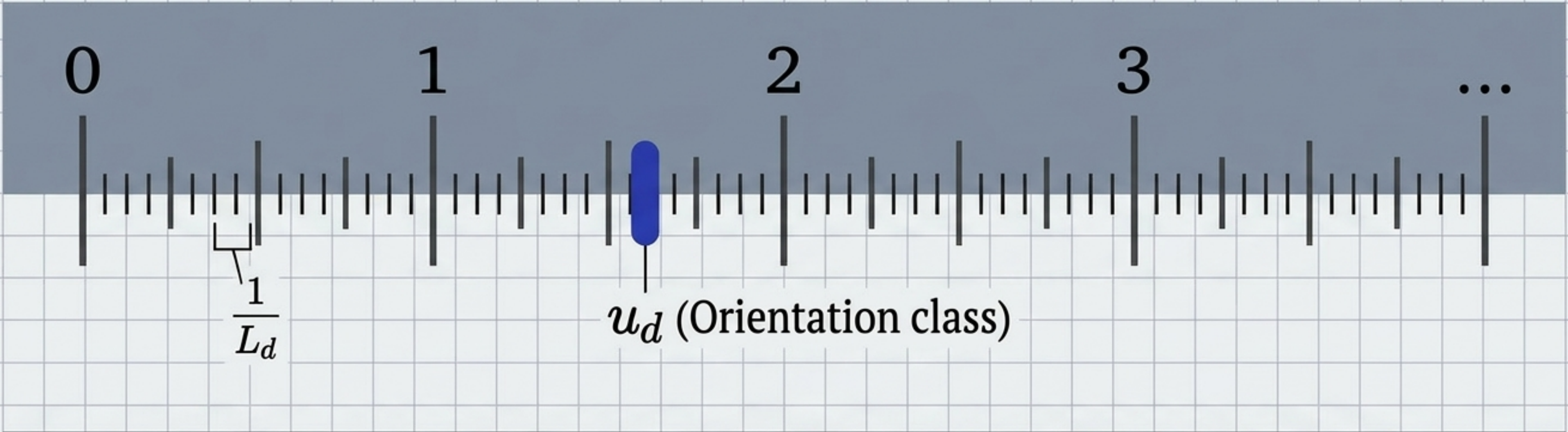
Theorem 1.1 (The Hilbert-Smith Conjecture)

Let $n \geq 1$. A locally compact second-countable Hausdorff group acting faithfully and jointly continuously on a connected Hausdorff second-countable topological n -manifold without boundary is a Lie group.

Theorem 1.2 (p -adic exclusion)

Every jointly continuous action of $\mathbb{Z}_p$ on such an n -manifold has a nonzero kernel (it cannot act faithfully).

Proof Step 1: Establish a fixed numerical lattice on an odd sphere.

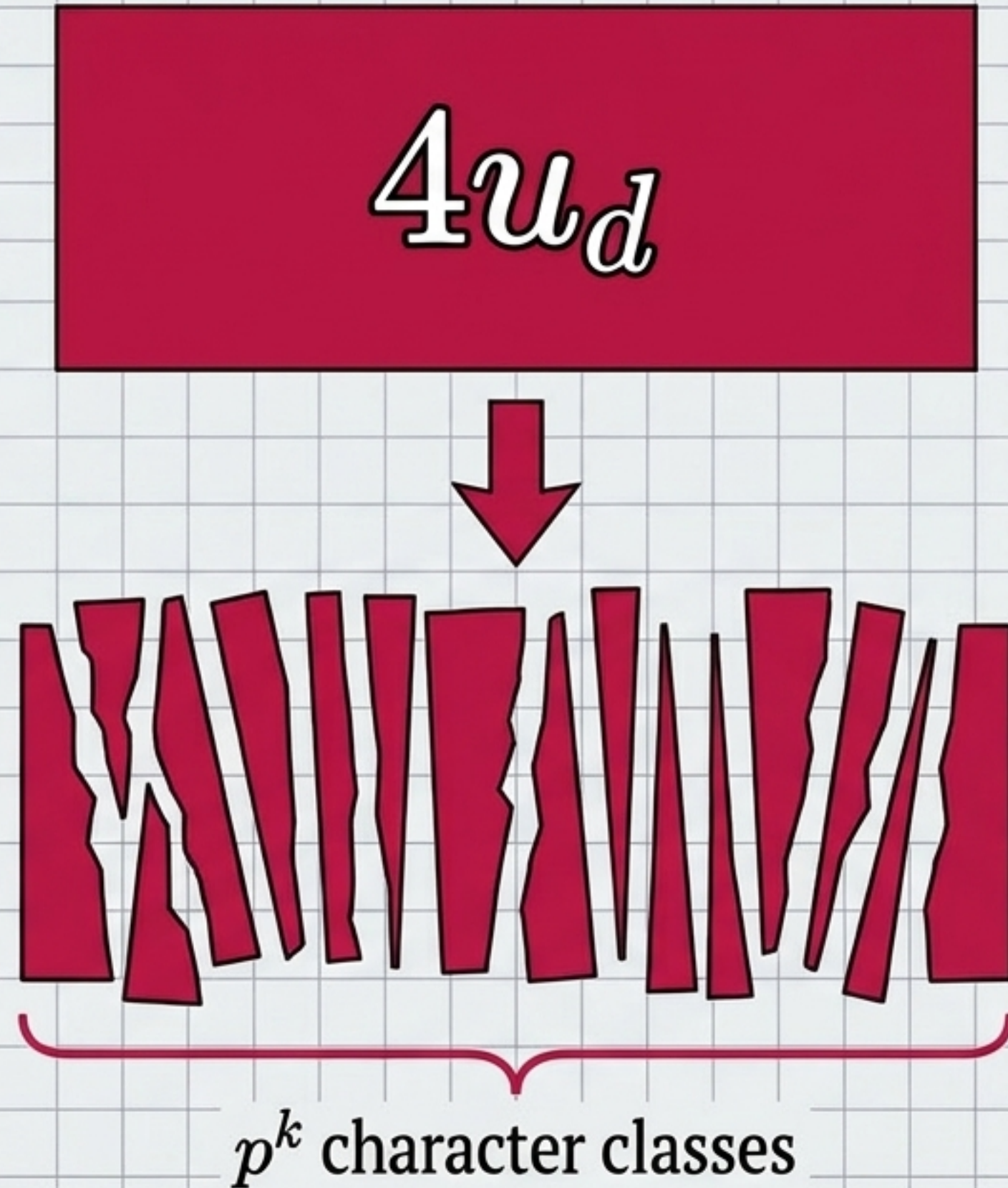


The Invariant	The Lattice	The Crucial Detail
Construct a Witt-group signature invariant on an odd sphere S^d (where $d > n$).	The rational images of these classes lie on a 1D line, specifically constrained to the lattice $(1/L_d)\mathbb{Z}u_d$.	L_d is a power of 2 that is permanently fixed before any character choices (k) are made.

Proof Step 2: A faithful action generates pieces that sum to $4u_d$.

The Split

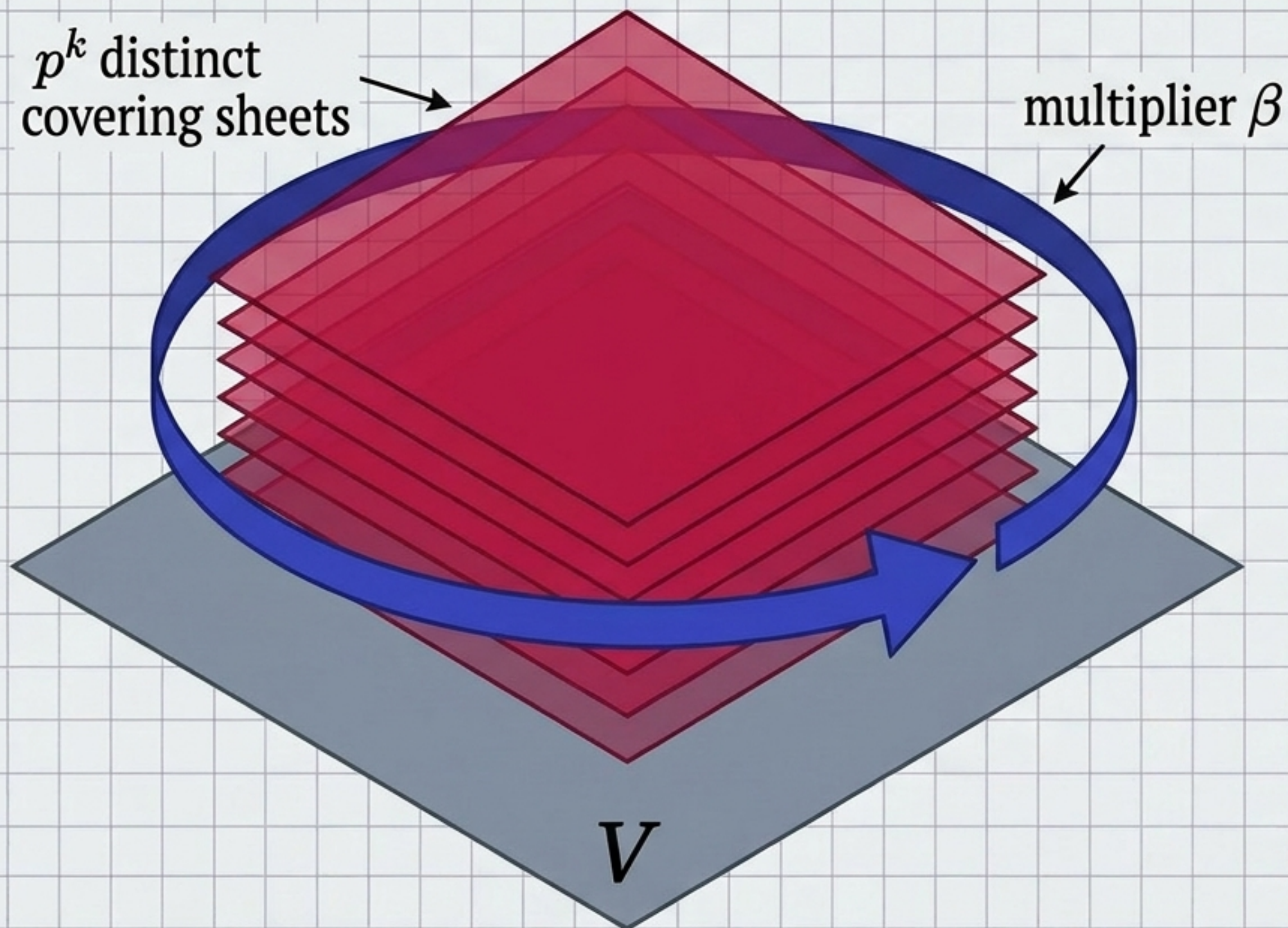
Assuming a faithful action exists, it produces p^k **distinct integral character classes** for any $k \geq 1$.



The Sum

No matter how many pieces (p^k) the action generates, their total sum must perfectly equal the original $4u_d$ block.

Proof Step 3: Over a specific region, local symmetry forces all pieces to be identical.



Covering Sheets:

The faithful action guarantees a region V with p^k separate covering sheets.

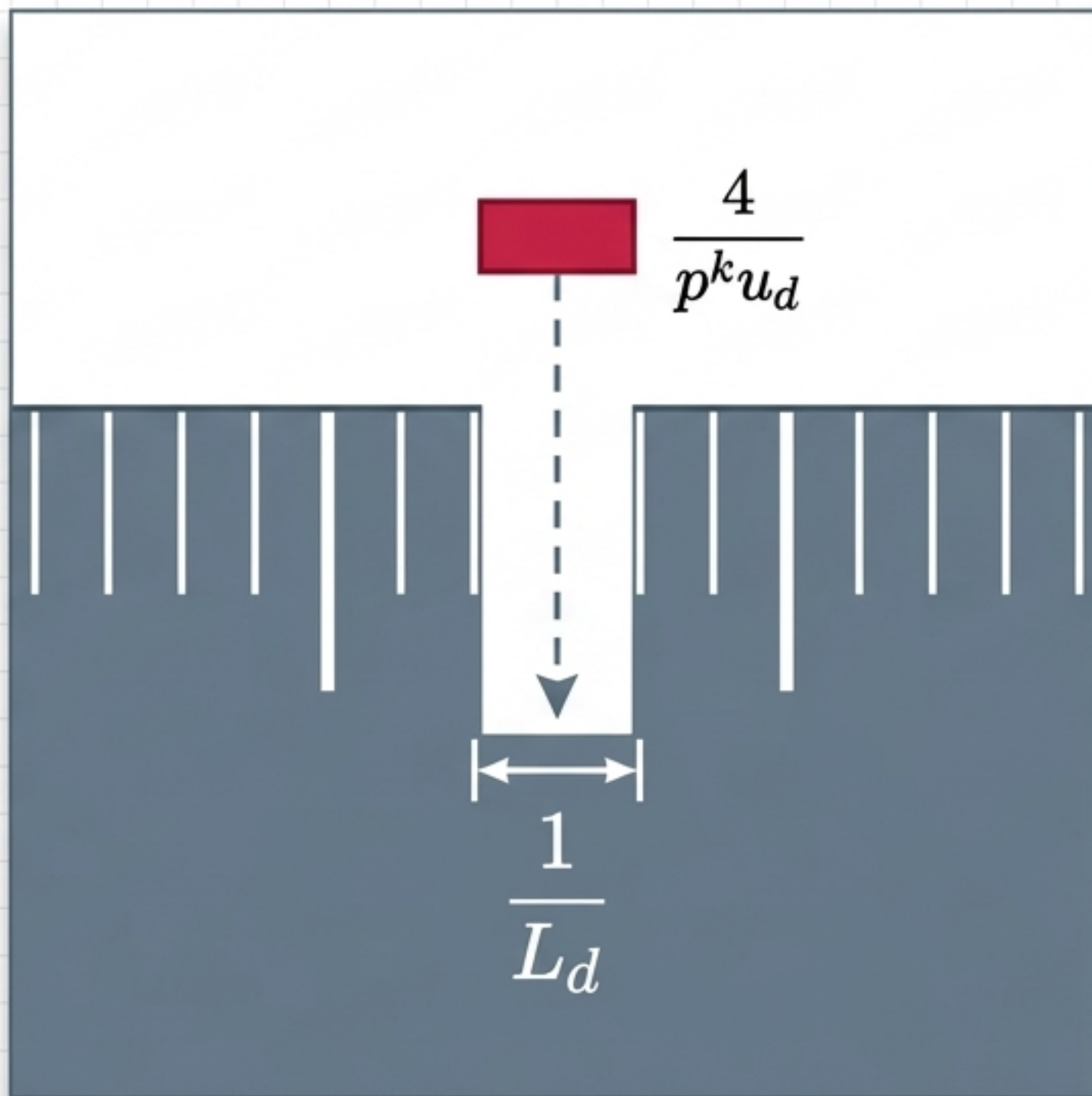
Cyclic Permutation: A locally constant multiplier β rotates the character parts across these sheets.

The Consequence: For any mathematical tests concentrated entirely over V , these character classes are forced to be perfectly equal.

Proof Step 4: Serre's odd-sphere theorem triggers the contradiction.

Moving the Test

Serre's theorem provides a degree- s map that moves our test entirely into region V .



The Trap Snaps

Because they are in V , the pieces must be equal (each is exactly $4/p^k u_d$).

We can choose k so large that $p^k > 4L_d$.

The piece is now smaller than the fixed lattice, making it mathematically impossible.

Epistemic Status: A boundary-pushing, yet unverified artifact.



AI-Generated

An unrefereed preprint produced by OpenAI (Sep 2026).



No Formalization

The proof lacks Lean formalization or machine verification. It relies on human-readable mathematical logic.



Scope Limits

Explicitly stated for manifolds without boundary and second-countable groups.

From a 1900 mystery to a digital lattice contradiction.

- **Synthesis:** The Hilbert-Smith conjecture survived a century because it required bridging extreme geometric flexibility with rigid algebraic topology.
- **The AI Approach:** This proof sidesteps geometric regularity **entirely**, using a Witt-group lattice to force an unavoidable numerical contradiction on the totally disconnected p-adic integers.
- **Takeaway:** Whether this proof withstands peer review or not, it represents a new era of mathematics—where AI synthesizes deeply disparate topological tools to break century-old deadlocks.